

Action-Compatible Failure-Aware Retrieval for VLA Inference Caching

TRACER refinement note

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Abstract

This note rewrites the current TRACER retrieval idea report as a concrete method proposal with explicit formulas. The refined thesis is that cross-trajectory cache retrieval for staged VLA inference needs three coupled decisions: what representation makes two entries reuse-compatible, whether the query lies closer to success support or failure support, and how much temporal chain context should be compared at the current phase. The method therefore combines outcome-compatible retrieval projections, failure-aware dual retrieval, and dynamic chain-depth scoring inside the existing TRACER full-hit / warm-start / miss interface.

1 Problem Anchor

Bottom-line problem. TRACER must reduce staged vision-language-action (VLA) inference work by reusing cross-trajectory cached state while preserving task success close to pure inference.

Must-solve bottleneck. The current retrieval mechanism ranks candidates by observation-side similarity using hand-designed modality fusion and fixed chain context. This can retrieve candidates that look plausible but are unsafe for action reuse or warm start. The current retrieval also uses fixed chain depth and positive-only support, so it cannot adapt temporal context or detect that a query is near a known failure mode.

Non-goals. The goal is not to replace the host VLA with retrieval-as-control, not to train a new policy, and not to make vector indexing the main contribution. Indexing can be upgraded later without changing the retrieval semantics.

Constraints. The host VLA is frozen. Cache entries are built from existing successful and failed rollout traces. New trainable components must be small: lightweight projection heads, threshold calibration, and at most a shallow chain-depth selector. The method must preserve TRACER’s three-way verdict interface: full hit, warm start, or miss.

Success condition. The method succeeds if it improves the success/inference-ratio Pareto frontier over raw-feature, success-only, fixed-depth retrieval while reducing unsafe full hits, without making the cache behave like a standalone retrieval policy.

2 Technical Gap

Existing neighboring systems leave a gap. RT-Cache retrieves image-action snippets and replays them directly, but it has no quality-aware fallback path. VLA-Cache and related token-caching methods reuse computation within an episode, but they do not retrieve cross-trajectory action or denoise state. Speculative VLA methods verify drafted actions or tokens, but they do not define a failure-indexed cache boundary for cross-trajectory reuse.

The current TRACER retrieval mechanism can be summarized as

$$C_t = \text{Search}(\mathbf{q}_t, \mathcal{D}^+), \quad v_t = \text{Verdict}(C_t), \quad (1)$$

where $\mathcal{D}^+$ is a library of successful cache chains and v_t is one of $\{\text{hit}, \text{warm}, \text{miss}\}$. The missing object is not merely a better nearest-neighbor index. The missing object is a *support boundary*: a way to decide whether the query is near reusable success support or near known failure support, and a way to make the retrieval metric and chain depth reflect cache-reuse compatibility rather than raw observation similarity alone.

3 Method Thesis

One-sentence thesis. Project cache queries into an outcome-compatible space, score an adaptive chain segment against both success and failure support, and use the resulting support margin to gate full-hit and warm-start reuse.

Contribution 1: outcome-compatible retrieval. The first contribution is a small learned projection whose supervision is future action and warm-start compatibility, not policy imitation.

Contribution 2: failure-aware support boundary. The second contribution is dual retrieval over success and failure cache support, producing a margin that blocks unsafe reuse.

Contribution 3: dynamic temporal receptive field. The third contribution is a query-dependent chain-depth policy that chooses how much cross-trajectory temporal context to compare before Verdict.

4 Notation

At online step t , the cache query is a set of modality fields

$$\mathbf{q}_t = \{q_t^{(i)}\}_{i=1}^M, \quad (2)$$

where modalities may include camera streams, prompt fields, and robot state. A cache entry d stores

$$d = \left(\{e_d^{(i)}\}_{i=1}^M, \mathbf{a}_d, \{x_{d,\tau}\}_{\tau \in \mathcal{T}}, \text{step}(d), \text{prev}(d), \text{next}(d), y_d \right), \quad (3)$$

where $e_d^{(i)}$ is the stored modality field, $\mathbf{a}_d$ is the cached action chunk, $x_{d,\tau}$ is an optional partial denoise snapshot, $\text{step}(d)$ is the source-chain position, and $y_d \in \{+1, -1\}$ marks success or failure support. The positive and negative libraries are

$$\mathcal{D}^+ = \{d : y_d = +1\}, \quad \mathcal{D}^- = \{d : y_d = -1\}, \quad \mathcal{D} = \mathcal{D}^+ \cup \mathcal{D}^-. \quad (4)$$

5 Baseline Retrieval

For a query $\mathbf{q}_t$ and candidate d , TRACER computes per-modality ranks. For similarity-oriented modalities, lower rank means higher similarity:

$$\text{rank}_i(\mathbf{q}_t, d; \mathcal{D}) = 1 + \sum_{d' \in \mathcal{D}} \mathbb{I} \left[\text{sim}_i(q_t^{(i)}, e_{d'}^{(i)}) > \text{sim}_i(q_t^{(i)}, e_d^{(i)}) \right]. \quad (5)$$

For robot state fields based on distance, the inequality is reversed or the distance is negated before ranking. Weighted reciprocal rank fusion is

$$\text{RRF}(\mathbf{q}_t, d; \mathcal{D}) = \sum_{i=1}^M \frac{w_i}{k + \text{rank}_i(\mathbf{q}_t, d; \mathcal{D})}, \quad w_i \geq 0, \quad \sum_i w_i = 1. \quad (6)$$

For chain-aware retrieval with depth T , the score of an anchor d is

$$S_T(\mathbf{q}_{t:t-T}, d; \mathcal{D}) = \sum_{\ell=0}^T \alpha_\ell \text{RRF}(\mathbf{q}_{t-\ell}, d^{-\ell}; \mathcal{D}), \quad \alpha_\ell \geq 0, \quad \sum_{\ell=0}^T \alpha_\ell = 1, \quad (7)$$

where $d^{-\ell}$ is the ℓ -step predecessor of d . Anchors whose predecessor links cannot fill the requested depth are excluded from the depth- T candidate set.

6 Mechanism 1: Outcome-Compatible Retrieval Projection

The first mechanism fixes the meaning of nearest neighbor. Raw observation features can make two states look close even when their next actions or warm-start states are incompatible. Each modality field is therefore projected by a small head

$$z_t^{(i)} = h_{\theta_i}(q_t^{(i)}), \quad z_d^{(i)} = h_{\theta_i}(e_d^{(i)}). \quad (8)$$

The projected similarity replaces the raw similarity inside Equation (6):

$$\text{sim}_i(q_t^{(i)}, e_d^{(i)}) = \frac{\langle z_t^{(i)}, z_d^{(i)} \rangle}{\|z_t^{(i)}\|_2 \|z_d^{(i)}\|_2}. \quad (9)$$

The heads are deliberately small and the host VLA remains frozen. The projection is not a new policy; it is a cache-reuse metric.

6.1 Compatibility Labels

For cache entries a and b , define an action-compatibility target

$$c_{ab}^A = \exp\left(-\frac{\|\mathbf{a}_{a,a+H} - \mathbf{a}_{b,b+H}\|_2^2}{\sigma_A^2}\right), \quad (10)$$

where $\mathbf{a}_{a,a+H}$ is the next- H action chunk. If denoise snapshots exist, define a warm-start compatibility target

$$c_{ab}^X = \max_{\tau \in \mathcal{T}} \exp\left(-\frac{\|x_{a,\tau} - x_{b,\tau}\|_2^2}{\sigma_X^2}\right). \quad (11)$$

The combined compatibility is

$$c_{ab} = \eta c_{ab}^A + (1 - \eta) c_{ab}^X, \quad \eta \in [0, 1]. \quad (12)$$

If denoise snapshots are unavailable, set $\eta = 1$. The most defensible novelty is the case $\eta < 1$, where the metric is tied to cache warm-start compatibility rather than generic action-sequence similarity.

6.2 Projection Loss

For anchor a , choose positives

$$\mathcal{P}(a) = \{b : c_{ab} \geq \rho_+, y_b = +1\} \quad (13)$$

and hard negatives

$$\mathcal{N}(a) = \{b : c_{ab} \leq \rho_-\} \cup \{b : y_b = -1\}. \quad (14)$$

An InfoNCE-style loss is

$$\mathcal{L}_{\text{proj}} = - \sum_a \log \frac{\sum_{p \in \mathcal{P}(a)} \exp(\text{sim}(z_a, z_p)/\tau)}{\sum_{p \in \mathcal{P}(a)} \exp(\text{sim}(z_a, z_p)/\tau) + \sum_{n \in \mathcal{N}(a)} \exp(\text{sim}(z_a, z_n)/\tau)}. \quad (15)$$

This should be presented as acceleration-aware representation learning: the objective is not to train a policy, but to increase the probability that top retrieved candidates are accepted by the cache Verdict.

7 Mechanism 2: Failure-Aware Dual Retrieval

7.1 Success and Failure Scores

Instead of retrieving only from $\mathcal{D}^+$, the method retrieves from both success and failure support:

$$s_t^+ = \max_{d \in \mathcal{D}^+} S_{T_t}(\mathbf{q}_{t:t-T_t}, d; \mathcal{D}^+), \quad d_t^+ = \arg \max_{d \in \mathcal{D}^+} S_{T_t}(\mathbf{q}_{t:t-T_t}, d; \mathcal{D}^+), \quad (16)$$

$$s_t^- = \max_{d \in \mathcal{D}^-} S_{T_t}(\mathbf{q}_{t:t-T_t}, d; \mathcal{D}^-), \quad d_t^- = \arg \max_{d \in \mathcal{D}^-} S_{T_t}(\mathbf{q}_{t:t-T_t}, d; \mathcal{D}^-). \quad (17)$$

The failure-aware support margin is

$$m_t = s_t^+ - \lambda s_t^-, \quad \lambda \geq 0. \quad (18)$$

A positive margin means the query is closer to reusable success support than to known failure support; a negative or small margin means cache reuse is unsafe.

7.2 Dual-Retrieval Verdict Gate

Let u_t be the ordinary TRACER kinematic-quality signal computed from the positive candidate d_t^+ . The retrieval-side gate is

$$g_t = \sigma(\beta_0 + \beta_1 m_t + \beta_2 u_t + \beta_3 \Delta_t^+), \quad (19)$$

where Δ_t^+ is a positive-candidate ambiguity margin such as

$$\Delta_t^+ = S_{T_t}(\mathbf{q}_{t:t-T_t}, d_{t,1}^+; \mathcal{D}^+) - S_{T_t}(\mathbf{q}_{t:t-T_t}, d_{t,2}^+; \mathcal{D}^+). \quad (20)$$

The final action-reuse decision is

$$v_t = \begin{cases} \text{hit}, & m_t \geq \tau_{\text{hit}} \wedge u_t \geq \gamma_{\text{hit}}, \\ \text{warm}, & m_t \geq \tau_{\text{warm}} \wedge \exists \tau \in \mathcal{T} : x_{d_t^+, \tau} \text{ exists}, \\ \text{miss}, & \text{otherwise.} \end{cases} \quad (21)$$

with $\tau_{\text{hit}} > \tau_{\text{warm}}$. Equation (21) keeps the host VLA in the loop: failure evidence never triggers replay from $\mathcal{D}^-$; it only downgrades or blocks reuse from $\mathcal{D}^+$.

8 Mechanism 3: Dynamic Chain Depth

Let $\mathcal{T}_{\text{chain}} = \{0, 3, 5, 8\}$ be the allowed chain depths. Define a local ambiguity feature vector

$$r_t = [R_K(\mathbf{q}_t), \Delta_t^+, s_t^-, \|\mathbf{a}_{t-1} - \mathbf{a}_{t-2}\|_2, \text{step}(t)], \quad (22)$$

where $R_K(\mathbf{q}_t)$ is a local density radius, for example the distance to the K -th positive neighbor. A small depth policy selects

$$T_t = \arg \max_{T \in \mathcal{T}_{\text{chain}}} [\psi_T^\top r_t - \mu \text{cost}(T)]. \quad (23)$$

The policy is deliberately simple, but it is part of the retrieval mechanism: every score in Equations (7)–(18) uses the selected T_t . Its role is to adapt the temporal receptive field of cross-trajectory retrieval: shorter contexts are allowed in low-density or recovery states, while longer contexts are preferred in dense, steady phases where temporal smoothness matters.

9 Training and Calibration

Data construction. Use successful rollouts to build $\mathcal{D}^+$. Use failed rollouts, near-miss rollouts, or artificially corrupted cache decisions to build $\mathcal{D}^-$. Each entry keeps the same TRACER payload fields as the current database.

Projection training. Train the projection heads with Equation (15). The host VLA and action expert stay frozen. The raw-feature variant is retained only as an ablation.

Margin calibration. Fit thresholds $\tau_{\text{hit}}, \tau_{\text{warm}}$ and λ on a held-out calibration set by minimizing

$$\mathcal{L}_{\text{cal}} = \text{BadHitRate} + c_{\text{miss}} \text{MissRate} + c_{\text{warm}} \text{WarmCost}, \quad (24)$$

subject to a minimum task-success constraint

$$\text{SuccessRate} \geq \text{SuccessRate}_{\text{base}} - \epsilon. \quad (25)$$

10 Inference Algorithm

For each online step t :

1. Build query fields $\mathbf{q}_t$ at the TRACER checkpoint.
2. Project fields with h_{θ_t} , or bypass this step only in the raw-feature ablation.
3. Select T_t using Equation (23), or use a fixed depth for the ablation.
4. Retrieve d_t^+ from $\mathcal{D}^+$ and d_t^- from $\mathcal{D}^-$ using Equation (7).
5. Compute the failure-aware margin m_t in Equation (18).
6. Apply the decision rule in Equation (21).
7. If $v_t = \text{hit}$, return $\mathbf{a}_{d_t^+}$; if $v_t = \text{warm}$, resume from a compatible snapshot $x_{d_t^+, \tau}$; otherwise run the full host VLA.

11 Claim-Driven Validation

Claim 1: failure-aware retrieval reduces unsafe cache reuse. Compare success-only TRACER against dual success/failure retrieval. Measure bad full-hit rate, task success, inference ratio, and false fallback rate. The decisive evidence is a reduction in bad full hits at matched success or matched inference ratio.

Claim 2: outcome-compatible projection improves candidate quality only when compatibility labels include cache-reuse signals. Compare raw RRF, action-only projection, and action-plus-denoise projection. Report Retrieval@ K action compatibility, Verdict acceptance rate, and downstream success/inference ratio. This is a necessity check: if action-only projection gives the same result, the denoise compatibility story is not needed.

Claim 3: adaptive chain depth. Compare fixed $T \in \{0, 3, 5, 8\}$, oracle per-query T , and the simple policy in Equation (23). The decisive evidence is that the learned or calibrated policy recovers a meaningful fraction of the oracle gain while avoiding extra misses or unstable switching.

12 Metrics

Let N be the number of evaluation steps. The inference ratio is

$$\text{IR} = \frac{1}{N} \sum_{t=1}^N [\mathbb{1}[v_t = \text{miss}] + c_{\text{warm}}(\tau_t) \mathbb{1}[v_t = \text{warm}]], \quad (26)$$

where $c_{\text{warm}}(\tau_t)$ is the remaining denoise cost after warm start. The bad full-hit rate is

$$\text{BHR} = \frac{\sum_t \mathbb{I}[v_t = \text{hit}] \mathbb{I}[\text{bad}(t)]}{\sum_t \mathbb{I}[v_t = \text{hit}] + 10^{-8}}. \quad (27)$$

The false fallback rate is

$$\text{FFR} = \frac{\sum_t \mathbb{I}[v_t = \text{miss}] \mathbb{I}[\text{safe_reuse}(t)]}{\sum_t \mathbb{I}[\text{safe_reuse}(t)] + 10^{-8}}. \quad (28)$$

13 Positioning

The paper should not claim that action-aligned metric learning is new. CLASS, CLAMP, and related work make that risky. The defensible novelty is the integrated retrieval objective: outcome-compatible projections make candidate distance reflect cache reuse, failure-indexed retrieval defines a support boundary for frozen VLA inference reuse, and dynamic chain depth adapts the temporal receptive field of the retrieved segment. The failure-aware boundary remains the clearest novelty, but the proposal now includes all three mechanisms as the retrieval method.

14 Minimal Implementation Plan

1. Extend cache entry metadata with $y_d \in \{+1, -1\}$ and separate success/failure pools.
2. Implement dual retrieval by running the existing RRF/chain scorer over $\mathcal{D}^+$ and $\mathcal{D}^-$.
3. Train projection heads using the compatibility labels in Equation (12) and loss in Equation (15).
4. Implement dual retrieval by running the projected RRF/chain scorer over $\mathcal{D}^+$ and $\mathcal{D}^-$.
5. Add the dynamic depth selector in Equation (23).
6. Add margin-based thresholds in Equation (21).
7. Run ablations that remove each mechanism: raw features instead of projections, success-only retrieval instead of dual retrieval, and fixed depth instead of dynamic depth.

15 Closest Work

Work	Overlap	Difference
RT-Cache	Cross-trajectory retrieval and snippet replay	No failure index, no frozen-VLA fallback, no warm-start Verdict.
CLASS / CLAMP / RS-CL	Action-supervised representation learning	Train representations for control or VLA regularization, not cache-reuse compatibility for frozen inference acceleration.
VLA-Cache / adaptive token caching	VLA inference acceleration	Within-episode token reuse, not cross-trajectory action or denoise-state retrieval.
HeiSD / KERV	Kinematic-aware speculative VLA decoding	Autoregressive draft verification, not failure-indexed cache support for flow-style warm starts.
Failure-aware demonstration learning	Uses failed trajectories for safety	Uses failures for learning/value shaping, not inference-time dual retrieval fallback.